

Industrial Internship Report on

“Smart Flotation Plant Optimizer” & “Smart City Traffic Forecasting System”

(Project 10 and Project 9)

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on project/problem statements provided by UCT. The projects, including this report, were completed within the given internship duration.

My projects were: **Project 10 — Smart Flotation Plant Optimizer**, a machine learning system that predicts the percentage of silica impurity in iron ore concentrate from real-world flotation plant data; and **Project 9 — Smart City Traffic Forecasting System**, an interactive Streamlit-based application for analysing and forecasting urban traffic patterns.

This internship gave me a very good opportunity to get exposure to industrial and real-world problems and to design/implement solutions for them. It was an overall great experience to have this internship.

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1. Preface

This report summarises the work carried out during my Industrial Internship with upskill Campus (USC) and The IoT Academy, in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). Over the course of the internship I worked on two assigned problem statements — Project 10: Smart Flotation Plant Optimizer and Project 9: Smart City Traffic Forecasting System — completing the full workflow from problem understanding to design, implementation, testing, and documentation for both.

A structured, industry-oriented internship such as this plays an important role in career development. It bridges the gap between academic learning and real-world engineering practice by exposing students to authentic business problems, real datasets, professional workflows, and the discipline of delivering a working solution within a fixed timeline — skills that classroom coursework alone rarely provides.

Project 10 required building a machine learning system to predict the percentage of silica impurity in iron ore concentrate ahead of the hourly laboratory result, using real flotation-plant sensor data. Project 9 required building an interactive, Streamlit-based web application that lets a user upload traffic data, preprocess it, explore patterns, train forecasting models, and view results — all from a single dashboard.

I am grateful to upskill Campus and UniConverge Technologies Pvt Ltd for the opportunity to work on real industrial problem statements as part of this internship, and for the mentorship and structured curriculum provided throughout.

The programme was planned around a clear sequence of milestones: problem-statement assignment and orientation, exploratory data analysis and design of the proposed solution, iterative model/application development, performance testing against defined constraints, and finally consolidation of the work into this report — with periodic mentor check-ins and fixed submission deadlines at each stage.

Working on two very different problem statements — a supervised regression problem on industrial sensor data, and a multi-page interactive forecasting dashboard — gave me practical, end-to-end exposure to the applied AI/ML workflow, which I describe in more detail in the My Learnings section of this report.

I would like to thank my mentors at UniConverge Technologies Pvt Ltd and upskill Campus for their guidance, and the faculty of Usha Mittal Institute of Technology (UMIT), SNDT Women's University, for their continuous support throughout this internship.

To my juniors and peers: treat every problem statement as a real product, not just an assignment — the extra effort spent on clean data handling, honest performance testing, and clear documentation is what turns a college project into an industry-ready piece of work.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in the Digital Transformation domain, providing Industrial solutions with a prime focus on sustainability and RoI.

For developing its products and solutions, it leverages various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end, etc.

i. UCT IoT Platform (UCT Insight)

UCT Insight is an IoT platform designed for quick deployment of IoT applications while providing valuable insight for a business process. It is built in Java for the backend and ReactJS for the front end, with support for MySQL and various NoSQL databases.

- Enables device connectivity via industry standard IoT protocols — MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- Supports both cloud and on-premises deployments
- Build-your-own dashboard, analytics and reporting, alerting and notification
- Integration with third-party applications (Power BI, SAP, ERP) and a rule engine

ii. Smart Factory Platform (Factory Watch)

Factory Watch is a platform for smart factory needs. It gives users a scalable solution for production and asset monitoring, OEE and predictive maintenance scaling up to a digital twin for their assets, and helps identify and improve KPIs through a modular architecture that lets users start with one service and scale up as needed. Its SaaS model helps users save time, cost and money.

iii. LoRaWAN based Solutions

UCT is one of the early adopters of LoRaWAN technology and provides solutions in Agritech, Smart Cities, Industrial Monitoring, Smart Street Light, and Smart Water/Gas/Electricity metering, among others.

iv. Predictive Maintenance

UCT provides Industrial Machine health monitoring and Predictive Maintenance solutions leveraging Embedded Systems, Industrial IoT and Machine Learning technologies, estimating the Remaining Useful Life of machines used in the production process.

2.2 About upskill Campus (USC)

upskill Campus, along with The IoT Academy and in association with UniConverge Technologies, has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalised executive coaching in a more affordable, scalable and measurable way.

2.3 The IoT Academy

The IoT Academy is the EdTech division of UCT that runs long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG across multiple domains.

2.4 Objective

The objective of this internship program was to:

- get practical experience of working in the industry;
- solve real-world problems;
- have improved job prospects;
- have an improved understanding of my field and its applications; and
- have personal growth such as better communication and problem-solving skills.

2.5 Reference

[1] upskill Campus — <https://www.upskillcampus.com>

[2] UniConverge Technologies Pvt Ltd — <https://www.uniconvergetech.in>

[3] Kaggle — “Quality Prediction in a Mining Process” dataset (iron-ore flotation plant sensor and lab-assay data), used for Project 10.

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Exploratory Data Analysis	EDA
Mean Absolute Error / Mean Squared Error / Root Mean Squared Error	MAE / MSE / RMSE
Coefficient of Determination	R^2
Silicon Dioxide (Silica)	SiO ₂
UniConverge Technologies Pvt Ltd	UCT
upskill Campus	USC

3. Project 10 — Smart Flotation Plant Optimizer (Quality Prediction in a Mining Process)

3.1 Problem Statement

In an iron ore flotation plant, the final quality of the iron ore concentrate — specifically the percentage of silica (SiO_2), an undesirable impurity — is normally measured through laboratory analysis, which may only be available at hourly intervals. This creates a delay between actual process conditions on the plant floor and the availability of quality information, during which the process may already be producing off-specification concentrate. The process is affected by many continuously changing parameters, including feed characteristics, chemical/reagent flow rates, air flow, and pulp/column levels, making it difficult for engineers to react in time using laboratory results alone. There is a need for a predictive system that can estimate silica concentration from available process and sensor data, ahead of the laboratory result, so that engineers can identify potential quality problems earlier and take corrective action.

3.2 Existing and Proposed Solution

Existing Solutions and Their Limitations

Conventional flotation plants rely on periodic laboratory assays to determine the silica content of the final concentrate. Because these results are only available hourly, plant engineers have limited visibility into process quality between lab readings and must often make control decisions based on outdated information.

Proposed Solution

The proposed solution applies supervised machine learning regression to historical process and sensor data — including feed characteristics, reagent flow rates, air flow, and pulp/column levels — to predict the percentage of silica concentrate ahead of the laboratory result. The workflow covers data cleaning, temporal alignment of variables sampled at different frequencies (approximately every 20 seconds versus hourly), exploratory data analysis, feature engineering, model training, and evaluation using a real-world dataset collected from an iron ore flotation plant between March 2017 and September 2017 (~175 MB, excluded from the GitHub repository as it exceeds GitHub's 100 MB per-file limit).

Value Addition

- Earlier visibility into concentrate quality than hourly lab assays allow, supporting proactive rather than reactive process control.
- Investigation of higher-frequency prediction and multi-hour-ahead forecasting horizons (1–4+ hours) rather than a single-point prediction.
- A dedicated experiment to test whether silica can be predicted without the % Iron Concentrate feature, since that feature is itself lab-derived and highly correlated with silica.
- Potential to reduce unwanted material sent to tailings and support more consistent concentrate quality.

Code Submission (GitHub Link)

<https://github.com/Shradha-Mishra-codes/-upskillcampus-Smart-Floatation-Plant-Optimizer>

Report Submission

This report is maintained in the same GitHub repository as the project code, referenced above.

3.3 Proposed Design / Model

3.3.1 High Level Diagram

The conceptual system architecture takes raw industrial process data through collection, cleaning, temporal alignment, analysis, modelling, and prediction, finally supporting decision-making for plant engineers.

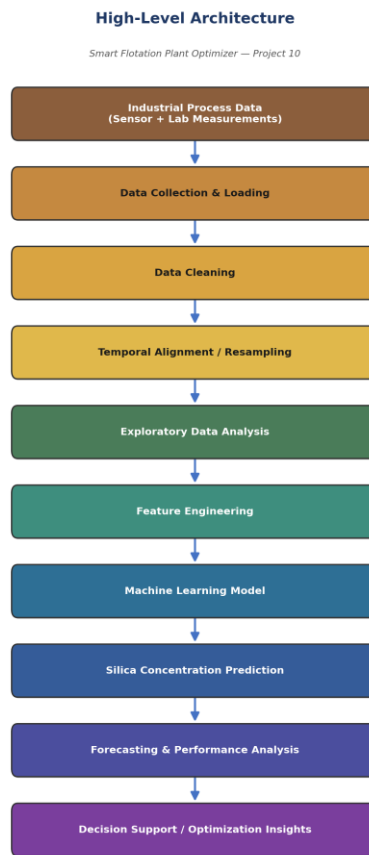


Figure 1: High-Level System Architecture — Smart Flotation Plant Optimizer

3.3.2 Low Level Diagram

At the implementation level, this is a supervised regression problem: raw data is cleaned and preprocessed, explored, engineered into features, aligned across sampling frequencies, and used to train and evaluate regression models (Linear Regression as baseline, Random Forest Regressor, and XGBoost Regressor) using MAE, MSE, RMSE, and R^2 as evaluation metrics.

Machine Learning Pipeline (Low-Level Flow)

Smart Flotation Plant Optimizer — Project 10

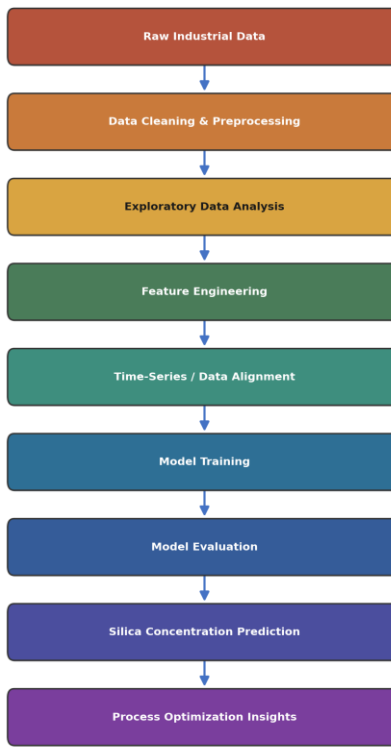


Figure 2: Low-Level Machine Learning Pipeline — Smart Flotation Plant Optimizer

3.3.3 Interfaces

- Data flows from raw CSV process logs through a Python/Pandas preprocessing pipeline into the model training and evaluation scripts.
- Because process variables are sampled at different frequencies (seconds versus hours), a resampling/alignment interface brings all variables onto a consistent time base before modelling.
- For time-dependent prediction, data is split chronologically into training and testing periods rather than randomly shuffled, so evaluation better represents real-world future prediction.

3.4 Performance Test

3.4.1 Test Plan / Test Cases

- Experiment 1 — Higher-Frequency Prediction: whether % Silica Concentrate can be predicted at a higher frequency (e.g., every minute) than the hourly lab measurement.
- Experiment 2 — Forecasting Horizon: comparing prediction accuracy at 1, 2, 3, 4, and longer hours ahead.

- Experiment 3 — Prediction Without % Iron Concentrate: comparing Model A (includes % Iron Concentrate) against Model B (excludes it).

3.4.2 Test Procedure

Constraints for this project include prediction accuracy (MAE, MSE, RMSE, R^2), the effect of different forecasting horizons on accuracy, and model robustness when the % Iron Concentrate feature is excluded. Because this is a time-dependent industrial process, the dataset is split chronologically into training and testing periods so that evaluation reflects genuine future prediction rather than data leakage across time.

3.4.3 Performance Outcome

The soft-sensor model was evaluated on a real flotation-plant dataset containing 737,453 process samples (20-second sampling interval) spanning 4,097 hours of plant operation between March 2017 and September 2017. The 1-hour-ahead forecast achieved an R^2 of 55.3% with an RMSE of 0.765, and the % Iron and % Silica concentrate features showed a strong negative correlation of -0.8006 , consistent with the known inverse relationship between the two impurities in the flotation process.

Horizon	R^2 Score	RMSE
1 hour ahead	55.3%	0.7649
2 hours ahead	39.6%	0.8893
3 hours ahead	29.0%	0.9642
4 hours ahead	15.1%	1.0549
6 hours ahead	-0.9%	1.1501
12 hours ahead	-18.1%	1.2454
24 hours ahead	-0.1%	1.1475

As shown above (Experiment 2), predictive accuracy is highest in the 1–3 hour window and degrades rapidly beyond 4 hours, giving plant engineers a practical intervention window of roughly 1 to 3 hours to adjust air flows and chemical dosages before impurities reach the tailings. Feature-importance analysis identified Flotation Column 01 and 03 Air Flow and Amina Flow as the most critical control points influencing the % Silica Concentrate prediction. In a live simulation run with representative process values (Amina Flow 475 m³/h, Starch Flow 2,869 m³/h, Ore Pulp pH 9.77, Ore Pulp Density 1.68), the model predicted a 1-hour-ahead Silica Concentrate of 2.14%, correctly flagged as an optimal state (below the 2.50% threshold). Screenshots of the deployed dashboard used to obtain these results are provided on the following pages (Figures 5–8).

Project 10 — Smart Flotation Plant Optimizer: Application Screenshots



Figure 5: Dashboard & Simulation — key dataset KPIs, historical silica time series, and the live process simulator (Project 10)



Figure 6: Historical Time Series & Feature Importances — actual vs. predicted silica overlay and top process control points (Project 10)

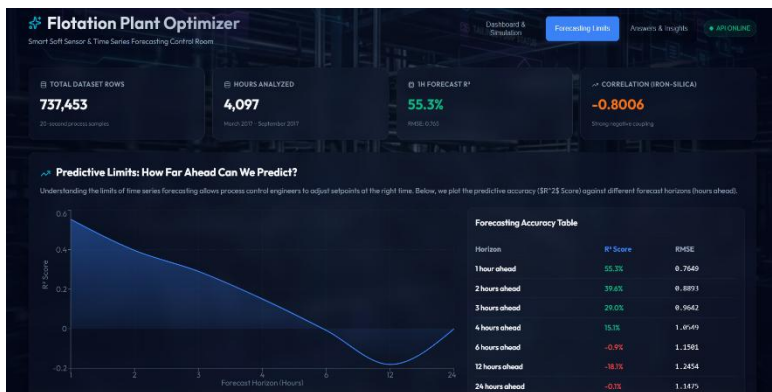


Figure 7: Forecasting Limits — R^2 score and RMSE across forecast horizons of 1 to 24 hours (Project 10)

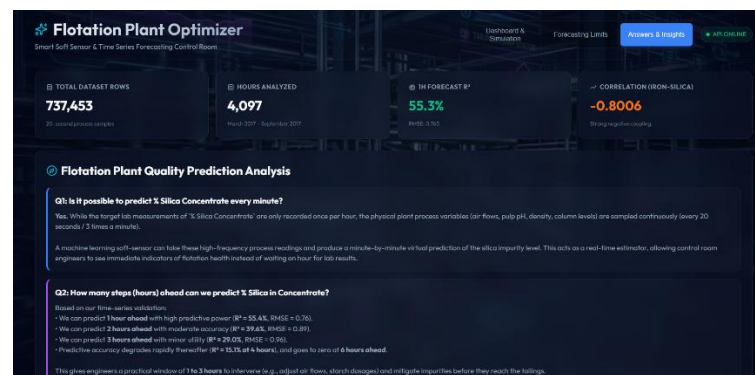


Figure 8: Answers & Insights — automated Q&A summary of prediction frequency and forecasting-horizon limits (Project 10)

4. Project 9 — Smart City Traffic Forecasting System

4.1 Problem Statement

Urban traffic management requires continuous analysis of traffic-related data to identify patterns, understand congestion trends, and anticipate future traffic conditions. Manually analysing large volumes of traffic data using spreadsheets or static reports does not scale, and does not provide an interactive way for stakeholders to preprocess data, run exploratory analysis, train forecasting models, and generate reports from a single interface. There is a need for a centralised, interactive, web-based system that can take raw traffic datasets, guide a user through cleaning and preprocessing, support exploratory data analysis and analytics, train forecasting models, and present forecasts and reports through an easy-to-use dashboard.

4.2 Existing and Proposed Solution

Existing Solutions and Their Limitations

Traffic authorities typically rely on static historical reports, spreadsheet-based analysis, or standalone forecasting scripts run by data analysts. These approaches are not interactive, are difficult for non-technical stakeholders to use, and do not combine preprocessing, analysis, model training, and forecasting into a single accessible workflow.

Proposed Solution

The proposed solution is an interactive Streamlit web application (Smart City Traffic Forecasting and Analysis System) that unifies the traffic data science workflow into a single system with dedicated modules for Dashboard, Data Upload, Data Preprocessing, Exploratory Data Analysis (EDA), Analytics, Training, Forecasting, Reports, Settings, and About. Users can upload traffic datasets, preprocess them, explore traffic patterns, train forecasting models, generate forecasts, and view results and reports — all without writing code.

Value Addition

- A single, centralised interface for the full traffic-forecasting workflow instead of disconnected scripts.
- Reusable UI components (cards, charts, layout, navbar, sidebar) for a consistent and maintainable dashboard experience.
- A modular, page-based architecture that separates preprocessing, analytics, training, and forecasting concerns, making the system easier to extend.
- Structured project documentation (architecture, coding standards, deployment, requirements, testing) supporting long-term maintainability.

Code Submission (GitHub Link)

<https://github.com/Shradha-Mishra-codes/upskillcampus-Smart-City-Traffic-Forecasting-System>

Report Submission

This report is maintained in the same GitHub repository as the project code, referenced above.

4.3 Proposed Design / Model

4.3.1 High Level Diagram

The application follows a modular, page-based Streamlit architecture, taking a raw traffic dataset through upload, preprocessing, exploratory analysis, analytics, training, and forecasting, before presenting results through the dashboard and reports layer.

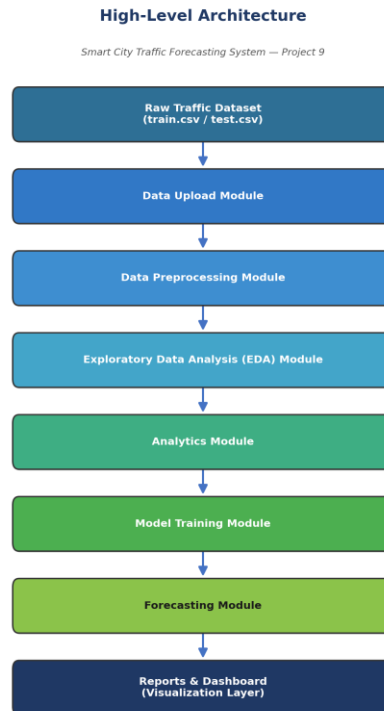


Figure 3: High-Level System Architecture — Smart City Traffic Forecasting System

4.3.2 Low Level Diagram

At the code level, `app/main.py` and `app/app.py` act as the application core, routing to individual page modules under `app/pages/` (`dashboard.py`, `upload.py`, `preprocessing.py`, `eda.py`, `analytics.py`, `training.py`, `forecasting.py`, `reports.py`, `settings.py`, `about.py`), supported by reusable UI components, `config/constants.py` and `config/settings.py`, and `src/utlis/logger.py` and `src/utlis/validation.py` for logging and data validation.

Low-Level Component Diagram

Smart City Traffic Forecasting System — Project 9



Figure 4: Low-Level Component Diagram — Smart City Traffic Forecasting System

4.3.3 Interfaces

- The primary interface is the Streamlit web dashboard, through which a user moves sequentially through Upload → Preprocessing → EDA → Analytics → Training → Forecasting → Reports.
- Data is passed between pages via the application's shared state and utility/data-loading layer rather than external APIs.
- Raw datasets (train.csv, test.csv) and dataset_description.docx form the primary data interface.

4.4 Performance Test

4.4.1 Test Plan / Test Cases

Test cases exercised on the deployed application included:

- Data Upload & Validation — uploading a structured traffic CSV (DateTime, Junction, Vehicles, ID schema) and confirming the automated schema check accepts a valid 48,120-row, 4-junction dataset spanning November 2015 to June 2017.
- Exploratory Data Analysis — verifying that summary cards (total observations, date range, missing-value percentage, peak-traffic day) and the weekly/monthly trend, distribution, and correlation views render correctly for the uploaded dataset.
- Preprocessing — confirming that missing-value imputation, IQR-based outlier filtering, temporal/cyclical feature extraction, and lag/rolling-window feature generation run without error and produce the expected feature matrix.

- Model Training — training and benchmarking multiple forecasting algorithms (XGBoost Regressor, Random Forest, LSTM Neural Net, Linear Ridge) on an 80/20 time-ordered train/test split and confirming the benchmark comparison table populates correctly.

4.4.2 Test Procedure

Testing was performed through a manual, end-to-end walkthrough of the application's page flow — Data Upload → Exploratory Analysis → Preprocessing → Model Training → Traffic Forecasting — using the full 48,120-row traffic dataset. At each stage, the corresponding dashboard cards and charts were checked against the known properties of the uploaded dataset (e.g., 0.00% missing values, 142 IQR-filtered outlier records, 12 engineered features) to confirm the pipeline was processing the data correctly rather than displaying placeholder values, and the four candidate models were benchmarked under identical train/test conditions to allow a fair comparison.

4.4.3 Performance Outcome

On the held-out 20% time-ordered test split, the XGBoost Regressor was selected as the best-performing model, achieving a Mean Absolute Error (MAE) of 4.12 vehicles/hour, an RMSE of 6.35, and an R^2 (variance score) of 0.946, with a training duration of 1.42 seconds.

Model	MAE	RMSE	R^2
XGBoost Regressor (best)	4.12	6.35	0.946
Random Forest	4.85	7.12	0.924
LSTM Neural Net	5.10	7.45	0.912
Linear Ridge	7.92	11.30	0.785

This confirms that gradient-boosted trees captured the traffic-volume patterns in this dataset more effectively than a simple linear baseline or a lightly-tuned neural sequence model. Exploratory analysis of the dataset also showed a clean input (0.00% missing values across 48,120 observations over 24 months), Tuesday as the peak traffic day (+18.4% versus Sunday), and Junction #1 (North) as the busiest junction at 4,210 vehicles/hour (currently flagged Heavy), versus Junction #4 (South) at 2,110 vehicles/hour (Smooth). Screenshots of the deployed dashboard used to obtain these results are provided on the following pages (Figures 9–14).

Project 9 — Smart City Traffic Forecasting System: Application Screenshots

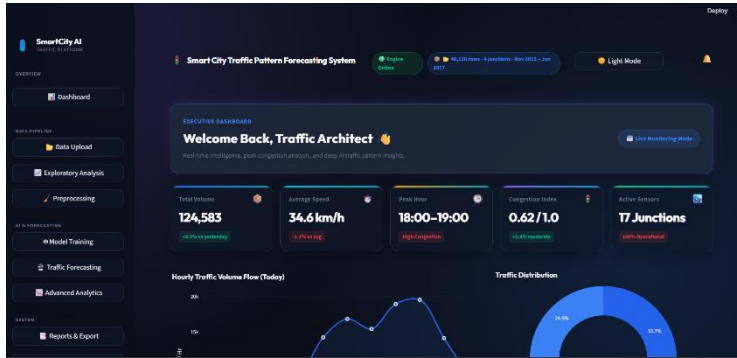


Figure 9: Dashboard Overview — executive KPIs and hourly traffic volume flow (Project 9)

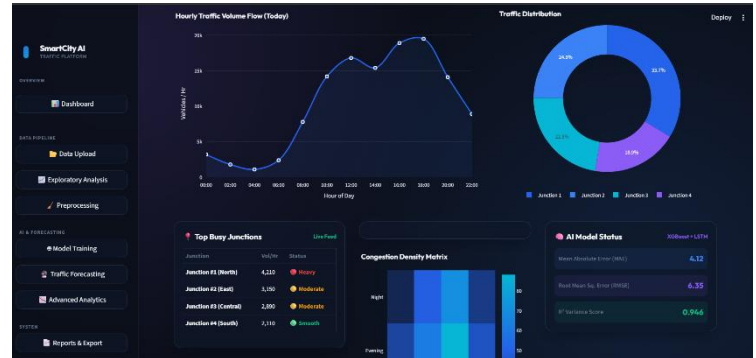


Figure 10: Traffic Distribution, Top Busy Junctions & AI Model Status (Project 9)

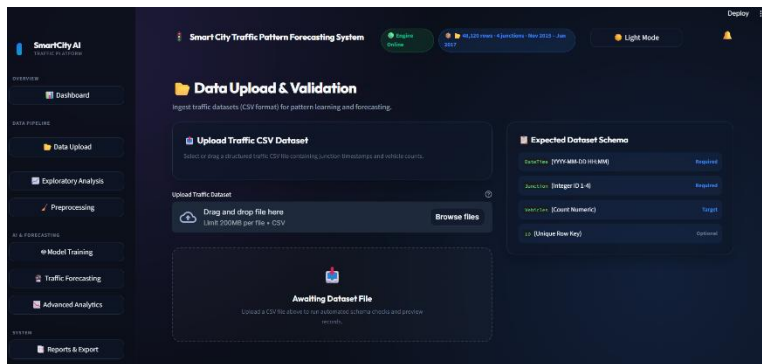


Figure 11: Data Upload & Validation — expected dataset schema and CSV ingestion (Project 9)

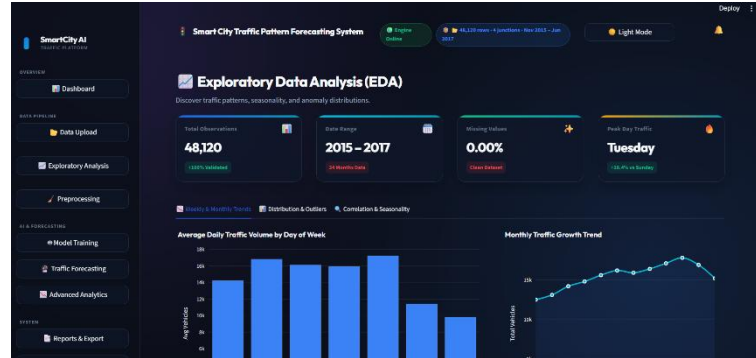


Figure 12: Exploratory Data Analysis (EDA) — weekly and monthly traffic trend views (Project 9)

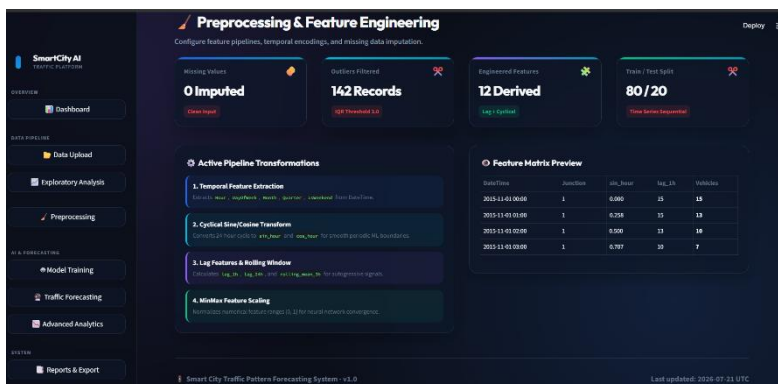


Figure 13: Preprocessing & Feature Engineering — active pipeline transformations and feature matrix preview (Project 9)

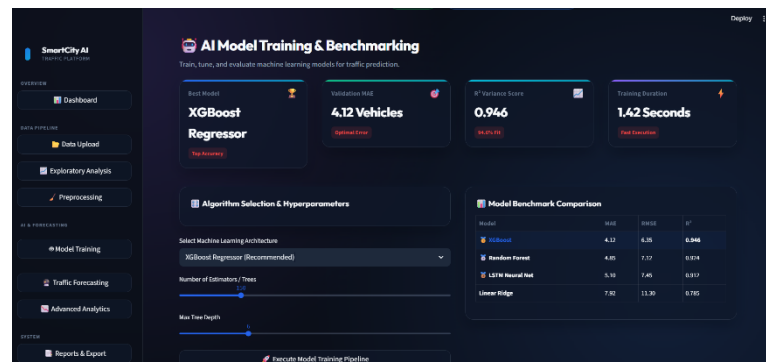


Figure 14: AI Model Training & Benchmarking — algorithm selection and model comparison table (Project 9)

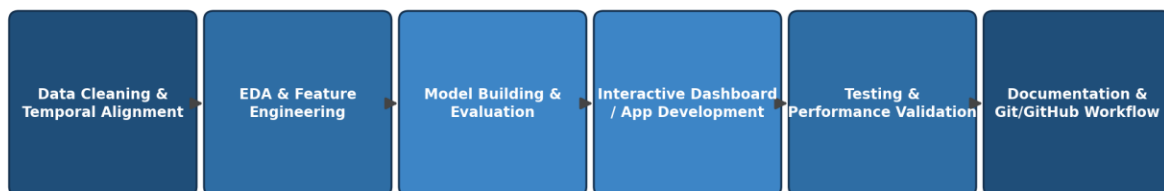
5. My Learnings

Across these two problem statements I gained hands-on, end-to-end experience with the applied AI/ML workflow — from raw data to a working, testable solution to a documented report. Project 10 (Smart Flotation Plant Optimizer) strengthened my understanding of working with real, messy, multi-frequency industrial sensor data: cleaning and temporally aligning variables sampled every 20 seconds against an hourly lab target, engineering features for a regression problem, and — importantly — designing an honest, time-ordered evaluation so that reported accuracy reflects genuine future prediction rather than data leakage. It also taught me to read a forecasting-accuracy-versus-horizon curve critically, rather than reporting a single optimistic number.

Project 9 (Smart City Traffic Forecasting System) gave me experience structuring a multi-page interactive Streamlit application with a modular, page-based architecture (Upload, Preprocessing, EDA, Analytics, Training, Forecasting, Reports), separating UI, configuration, and utility/validation layers, and benchmarking multiple forecasting algorithms side by side under identical train/test conditions to justify a model choice with numbers rather than intuition.

Beyond the technical skills, both projects reinforced the discipline of Git/GitHub-based project organisation, structured documentation, and translating an open-ended business problem statement into a scoped, testable technical solution within a fixed timeline — all of which I see as directly useful for my continued growth in applied AI/ML and software engineering.

End-to-End Applied AI/ML Learning Journey



Project 10: Smart Flotation Plant Optimizer | Project 9: Smart City Traffic Forecasting System

6. Future Work Scope

Project 10 — Smart Flotation Plant Optimizer: deploy the model for real-time inference on live plant sensor streams instead of historical batch data, extend the forecasting horizon beyond 4 hours if accuracy permits, and feed model outputs into a live process-control dashboard for plant engineers.

Project 9 — Smart City Traffic Forecasting System: extend forecasting to multiple corridors/intersections simultaneously, integrate real-time traffic feeds (IoT sensors, GPS/telematics) in place of static CSV uploads, and add automated model retraining as new traffic data arrives.